

Practice Problem Set -05 (Matrix-I)

Q1) Determine the rank of the following matrices.

a) $\begin{bmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 7 & 22 \end{bmatrix}$ Ans: $\rho(A) = 3$

b) $\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 13 \\ 1 & 3 & 4 & 1 \end{bmatrix}$ Ans: $\rho(A) = 2$

c) $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$ Ans: $\rho(A) = 3$

d) $\begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix}$ Ans: $\rho(A) = 2$

Q2. For what values of k the equations $x + y + z = 1$, $2x + y + 4z = k$, $4x + y + 10z = k^2$ have a solution and solve them completely in each case.

Ans: $k = 1, x = -3z, y = 2z + 1$; $k = 2, x = 1 - 3z, y = 2z$

Q3. Investigate for what values of λ and μ the simultaneous equations

$$x + y + z = 6, \quad x + 2y + 3z = 10, \quad x + 2y + \lambda z = \mu,$$

have (i) no solution, (ii) a unique solution, (iii) an infinite number of solutions.

Ans: (i) $\lambda = 3, \mu \neq 10$; (ii) $\lambda \neq 3, \mu \neq 10$; (iii) $\lambda = 3, \mu = 10$

Q4. Test the consistency and solve,

(i) $2x - 3y + 7z = 5, \quad 3x + y - 3z = 13, \quad 2x + 19y - 47z = 32.$

(ii) $x + 2y + z = 3, \quad 2x + 3y + 2z = 5, \quad 3x - 5y + 5z = 2, \quad 3x + 9y - z = 4$

(iii) $2x + 6y + 11 = 0, \quad 6x + 20y - 6z + 3 = 0, \quad 6y - 18z + 1 = 0$

(iv) $3x + 3y + 2z = 1, \quad x + 2y = 4, \quad 10y + 3z = -2, \quad 2x - 3y - z = 5.$

Ans: (i) Equations are inconsistent; (ii) Consistent; $x = -1, y = 1, z = 2$;

(iii) Equations are inconsistent; (iv) Consistent; $x = 2, y = 1, z = -4$.



Q5. Find the values of a and b for which the equations

$$x + ay + z = 3, \quad x + 2y + 2z = b, \quad x + 5y + 3z = 9$$

are consistent. When will these equations have a unique solution?

Ans: Case (i): If $a = -1, b = 6$, equations will be consistent and have infinite number of solutions;

Case (ii): If $a = -1, b \neq 6$, equations will be inconsistent;

Case (iii): If $a \neq -1, b$ has any value, equations will be consistent and have a unique solution

Q6. Show that if $\lambda \neq -5$, the system of equations

$$3x - y + 4z = 3, x + 2y - 3z = -2, 6x + 5y + \lambda z = -3, \text{ have a unique solution.}$$

If $\lambda = -5$, show that the equations are consistent. Determine the solutions in each case.

Ans: Case(i): $\lambda \neq -5, x = \frac{4}{7}; y = -\frac{9}{7}; z = 0$

Case(ii): $\lambda = -5, x = \frac{1}{7}(4 - 5k); y = \frac{1}{7}(13k - 9); z = k$ for all k